



HK IMO Training 2023-2024
Problem Set 4
For 28 October 2023



Problem 13. There are 5 points on the plane, with no three collinear. Some pairs of points are joined by a segment. We say that an ordered triple (A, B, C) of distinct points is good if one of the following holds:

- A and B are joined by a segment, and B and C are joined by a segment;
- A and B are not joined by a segment, and B and C are not joined by a segment.

Find the minimum number of good triples.

Problem 14. In $\triangle ABC$, the A -excircle touches the extensions of AB and AC at P and Q respectively. The B -excircle touches the extensions of BA and BC at S and T respectively. Let X and Y be the feet of perpendiculars from C to PQ and ST respectively. Prove that P, X, Y, S are concyclic.

Problem 15. Let $n > 1$ be a positive integer. Suppose $a_1, a_2, \dots, a_{2n}$ are integers with sum $n + 1$ such that $|a_j| \leq n$ for every j . Prove that some of $a_1, a_2, \dots, a_{2n}$ have sum 0.

Problem 16. Let $\{x\} = x - \lfloor x \rfloor$ denote the fractional part of a real number x . Prove that for any 20 consecutive positive integers, we can find one of them, say n , such that

$$k\sqrt{n} \{k\sqrt{n}\} > \frac{5}{2}$$

for all positive integers k .